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Beyond Black's equation: Full-chip EM/SM assessment in 3D IC stack

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ABSTRACT

Simulation flow for the die-scale assessment of interconnect stress-migration (SM) and electromigration (EM) with an accounted variation of residual stress was developed. Two complimentary methodologies based on finite-element sub-modeling and compact modeling were proposed. A novel EM and SM model which takes into account a vacancy exchange between grain boundaries and grain interior has demonstrated a capability of predicting times for void nucleation at different test conditions while avoiding unreliable assumptions used in the Black's equation-based assessment.

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1. Introduction

Continuous increase in the die size accompanying by reduction of the metal line cross sections and, hence, by increase of the current densities, which is governed by a technology scaling, results in an increasingly difficult EM signoff when the traditional EM checking approaches are employed. Widely predicted decrease in EM lifetime with the transition to advanced technological nodes is responsible for the pessimistic performance–reliability paradigm: high chip performance is accompanied inevitable by poor reliability and vice versa, a very reliable chip cannot demonstrate the top performance. EM-induced failure rates of the individual segment are considered as a measure of EM induced reliability and, in the extreme end, a mean time-to-failure (MTTF) of the weakest segment is accepted as a measure for the chip life-time. It results in a very conservative design rule for the current density that can be used in the chip design for a particular technology node in order to avoid EM failure.

A very different way to EM assessment can be proposed if we take a look at the interconnect reliability from the position of its functionality, when the failure of interconnect just means its inability to function properly [1]. There are two most important functions of the chip interconnect, which are: providing a connectivity between different parts of design for a proper signal propagating, and delivering a needed amount of voltage where it is required. While the cutting-off individual segments of the interconnect circuits can degrade both these functions, the role of EM is quite different in these two cases of degrading the power supply chain and the signal circuits. The difference is in the types of electrical currents employed in these two cases. Indeed, the signal lines carrying bidirectional or pulsed currents are characterized by very long times to the EM-induced failure. It is caused by a repetitive increase and decrease of the mechanical stress at the segment ends, caused by the atom accumulation and depletion due to interaction with the electron flow. In contrast, power lines carrying unidirectional currents can fail in much shorter time due to continues stress buildup under EM action. Thus, we can conclude that EM induced chip failure is happening when interconnect cannot deliver needed voltage to any gate of the circuitry. It means that loss of performance, which is a parametric failure, should be considered as the practical criterion of EM-induced failure rather than a catastrophic electrical breakdown or short. It is clear that a structure of the power grid, which is characterized by high level of redundancy, can affect the kinetics of failure development. Indeed, due to redundancy the failures of some amount of interconnect segments do not necessary result the voltage drop on the grid to become unacceptable [2]. Thus, more accurate and less pessimistic full-chip EM assessment and MTTF prediction will require development of new methods that deal with the grid structure and take redundancy into account. Well-developed percolation theory of grids and lattices should be employed. MTTF should be determined as an instant in time when a fraction of failed segments will increase the voltage drop above the acceptable level. Distributions of times-to-failure of individual segments should be provided by measurements.

Described EM assessment assumes a prior knowledge of current densities and temperatures in each segment across interconnect. A complexity of extraction of these distributions is acerbated by an uncertainty in work load taking place in modern chips. Its complex multi-modal behavior results a dependency of the power dissipated by different blocks on the modes of operation. It means that current densities and temperatures in different interconnect segments should be estimated for different workloads and should be

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used for prediction of MTTF happening in different scenarios including worst-case conditions for voltage drop [3].

Additional problem that should be addressed in order to develop a robust methodology for the full-chip EM assessment is an availability of the physics based models for void/hillock initiation and evolution that cause a time-dependent degradation of the segment electrical characteristics. Currently employed Blech limit [4] (for the out filtration of immortal segments) and Black's equation [5] (for calculating MTTFs for segments characterized by known current densities and temperatures) cannot handle an experimentally demonstrated dependency of MTTF on the residual stress [6,7]. Across-die variation of residual stress demands the Blech's "critical product" and Black's activation energy and current density exponent to be considered as layout dependent variables rather than experimentally determined constants. Hence, the new approach to the full-chip EM assessment should provide a robust methodology for EM failure check and capability for more optimistic prediction of the EM-restricted current density design rules for the future technological nodes. This target requires a development of new methodologies for accurate calculation of the worst case corner current densities and temperatures due to the existing workload uncertainties. A robust methodology for calculation of the across-interconnect distribution of residual stress caused by a sequence of the process steps and corresponding stress relaxations should be developed. Solid criterion of the failure on the basis of increasing voltage drop above acceptable level caused by EM-induced segment failures taking into account grid redundancy should be developed and validated with the chip performance measurements.

This paper describes a novel methodology for predicting the across-interconnect distribution of residual stress. Effect of residual stress on the EM-induced void initiation is analyzed. Developed methodology is demonstrated by simulation of stress generated in TSV-based 3D IC stack and its effect on EM-induced void nucleation.

2. Effect of TSV proximity on SM and EM interconnect reliability

Implementation of 3D IC through-silicon-vias (TSV)-based chip architecture is considered as the next step toward chip miniaturization and a further boost to chip performance. Thermo-mechanical reliability is a key concern for the adoption of the 3D IC technology. Indeed, the presence of extremely thin dies with thicknesses less than 50 μm, mounted on arrays of solder bumps, the presence of TSVs, i.e. large metal structures that are several tens or hundreds times larger than typical structures in BEoL stacks, taken together with operation conditions, which are characterized by temperatures and temperature gradients essentially higher and steeper compared to conventional planar chip architectures, inevitably generate mechanical stresses of values that significantly exceed the stresses in planar 2D dies. These stresses can be responsible for many reliability issues such as silicon cracking, TSV pop-up, delaminations in the back-end-of-line and back-chip-side redistribution (BEoL/BRDL) interconnects, etc., which have been discussed recently in many publications [8,9]. Stress is also associated with the degradation of the performance of active devices located in close proximity to TSV and TSV arrays, [10]. In addition to the mentioned reliability issues, the chip-package-interaction (CPI)-induced stress can exacerbate a variety of problems associated with materials aging phenomena such as electrical breakdown in inter-metal dielectric, EM, SM, etc. The last two phenomena are very sensitive to residual stress, since the critical stress that is needed for a failure initiation, is a combination of stresses generated by all possible sources. Hence, a capability of the across-die assessment of stresses generated by a variety of process steps is beneficial for chip design optimization as well as for generation of the critical design rules.

As an example let's consider stress evolution in the BEoL interconnect metal lines (Fig. 1), located in the close TSV proximity, caused by thermal and EM loads. In this simulation T_{ZS} = 573 K was taken as a zero stress temperature. Thermal ramp down from T_{ZS} to the test temperature of T_{test} = 373 K, and diffusion-induced stress relaxation at T_{test} result stress distributions shown in Fig. 2a and b. Fig. 2c shows the hydrostatic stress distribution in the same lines when the direct current (DC) was passing through TSV ($j = 8.8 \times 10^9$ A/m^2) and through the resolved two interconnect metal lines (shown in Fig. 1) ($j = 3.4 \times 10^{10}$ A/m^2) at T = 373 K. These results were obtained from solution of the system of second order partial differential equations, which comprise the linked multiphysics SM/EM model described in [11,12]. This model links the continuity equations, describing the intra segment evolution of vacancy and plated atom concentrations, with the force balance equation, describing the evolution of the stress components, and with the Laplace equation, in the case of EM load, for the current density distribution inside metal line. Simulation shows that slightly different values of the hydrostatic stresses were generated in these two lines, depending on their positioning relative to the TSV. It should be noted that the calculated void nucleation times, simulated for different current densities j, are approximated well by the power functions of j: $t_{nuc} \sim j^{-n}$. Void nucleation time was determined as the instance in time when the hydrostatic stress has reached for the first time the value of 500 MPa. It was found that the current density exponents "n" extracted from the simulation results were slightly different for these two lines. We obtained n = 1.88 for the line closer to the TSV and n = 1.84 for the further away one. The current densities employed in this experiment were spread in the interval of [4.9×10^9 A/m^2; 3.4×10^{10} A/m^2]. This result supports the fact of the dependency of "n" on stress. Dashed and solid lines in Fig. 2 show the comparison between distributions of hydrostatic stresses developed in similar line segments located nearby to and far away from the TSV. It is clear that the TSV presence changes the state of stress inside the nearby metal lines toward more tensile, which can be responsible for faster void nucleation.

2.1. Global stress modeling for die-scale SM & EM reliability assessment

Described above results were obtained for a single TSV embedded into thin silicon die. In reality, total distributions of strains and stresses inside thin tiers mounted on arrays of solder bumps and exposed to thermal impacts are governed not by TSVs only but by a combination of different stress sources such as warpage, bump-induced contact stresses, etc. As an example, Fig. 3a demonstrates distributions of a lateral stress inside a bottom tier of a stack, consisting of a substrate and two tiers. Fig. 3b shows distributions of the lateral stress along the line crossing the silicon bulk near the interface with interconnect above the array of flip-chip (FC) bumps.

Thus, in order to be able to predict the effect of residual stress on the SM/EM induced failures in the interconnect segments the global thermo-mechanical model for deformation of the whole stack should be solved. Similarly to the multi-scale simulation methodology, that was developed for the assessment of stress-induced device performance in 3D IC stacks [10], the reliability assessment also requires a development of the multi-scale simulation flow. The simulation methodology/flow should provide the capability for calculating across-interconnect variation of the mechanical stress generated by die stacking using the 3D IC TSV technology. Resolved stresses should be used as the initial-residual stresses in calculation of the degradation kinetics in interconnect

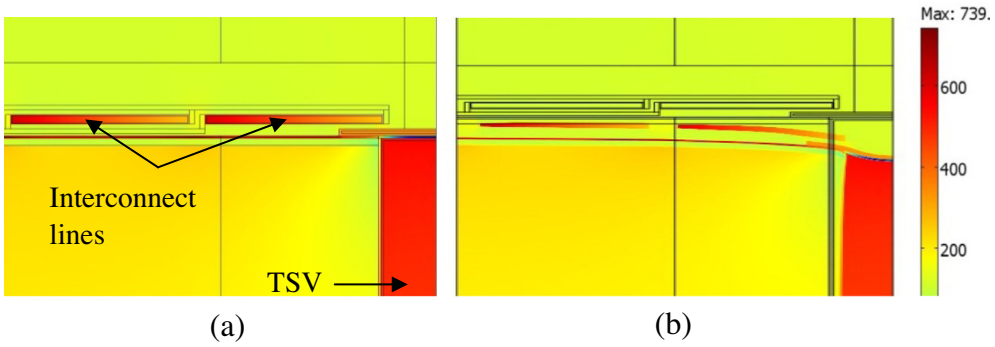


Fig. 1. BEoL interconnect metal lines in TSV proximity. Colour map shows the distribution of the hydrostatic stress caused by EM stressing and interaction with TSV (a), and deformations (b); deformed shape scale factors is 50 for (b).

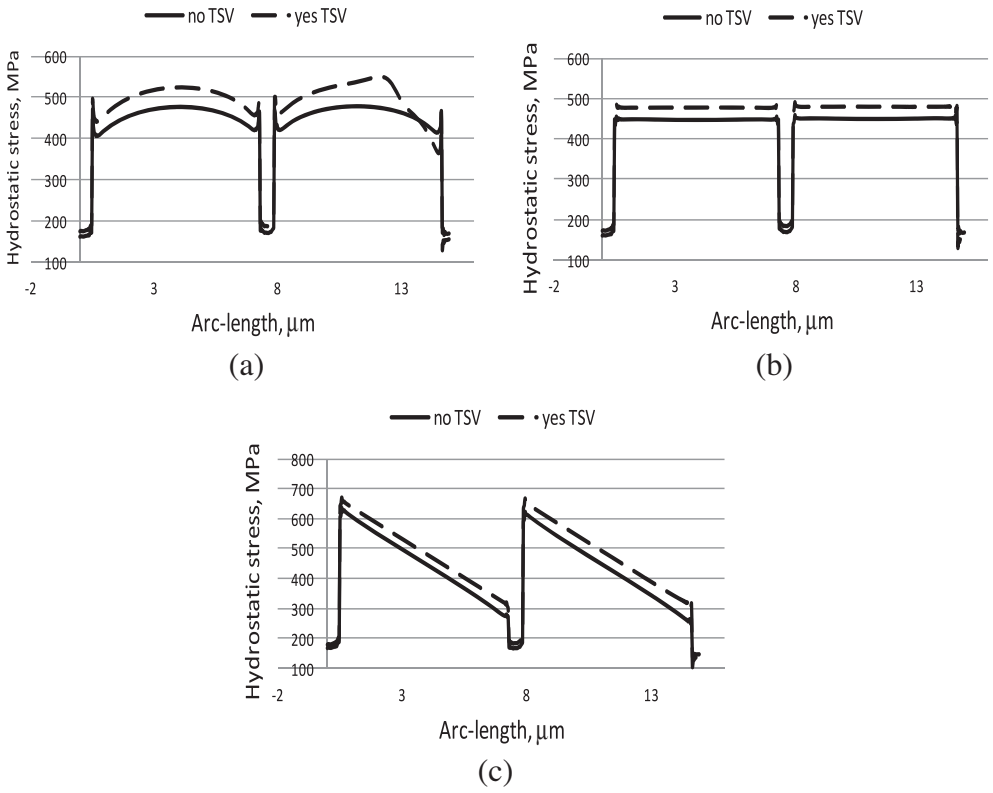


Fig. 2. Steady state distributions of the hydrostatic stress along two metal lines located close to TSV (dashed lines) and far away from TSV, when TSV induced stress is vanished (solid lines) at T_{test}, after the thermal ramp down from T_{ZS} to T_{test} (a), after completion of the diffusion-induced stress relaxation at T_{test} (b), and, finally, when the EM stressing was applied (c).

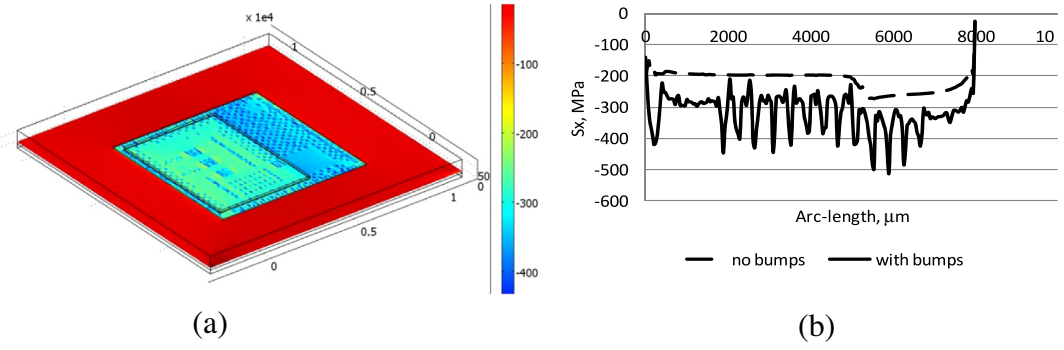


Fig. 3. Lateral stress distribution across the bottom die bulk (a); distributions of the lateral stress along the lines crossing the silicon above the array of FC bumps (b). Dashed line – uniform underfill layers were assumed, solid line – bumps were taken into account, (stress is in MPa).

metal segments that are subjects for EM failure. Employed models should be scalable for calibration at different technology nodes.

2.2. Die-scale SM & EM reliability assessment with FEA-based sub-modeling

Methodology for the multi-scale simulation of the distribution of stresses inside dies stacked in 3D structures, employing arrays of bumps and TSVs for the electrical interconnectivity, has been developed and described in number of papers (see for example [10] and cited literature there). It consists of FEA-based global-package-scale model for calculating fields of displacements at the interfaces of the dies, FEA-based die-scale model for calculating the strain propagation through BEoL and RDL interconnects to silicon, and, compact model-based stress relaxation governed by a die layout topology. In the considered case of SM & EM interconnect reliability, approximate stress/strain distributions inside interconnects are calculated at the die-scale with FEA-based simulation. At this step, interconnect is represented as a layer with non uniform, coordinate dependent mechanical properties. These properties (Young's modulus, Poisson factor, thermal extension coefficient, etc.) are derived based on theory of mechanical properties of anisotropic composites, [13], with the density, orientation and geometries of metal components extracted from the layout (GDSII or OASIS files) and technology files. Resolved stress distribution across interconnect displays an amplitude of stress variation that can be expected for SM/EM modeling. In order to resolve stress at the scale of individual metal line a sub-modeling technique should be employed. A small bin containing a line of interest should be cut from interconnect layer. Displacements on the bin faces, extracted from the die-scale simulation, should be used as boundary condition (BC). Same temperature load as used in the global and die-scale simulations should be employed. As a result the distributions of all stress components inside metal line will be available. Employment of this stress as a residual stress in the comprehensive SM & EM simulation model [12] allows to calculate stress evolution and to determine the critical times when stress reaches the critical value (MTTF). While being capable of providing the accurate results, this method cannot be considered as a feasible approach to the chip-scale EM assessment due to very large size of the model.

2.3. Die-scale SM & EM reliability assessment with compact models

Another approach to the chip-scale reliability assessment is based on compact modeling. A number of attempts to develop SM & EM compact model-based verification tools have been reported in the past, and even more attempts are undertaken currently [14,15]. Unfortunately, majority of the proposed approaches to the EM reliability assessment have suffered and continue to suffer the absence of the reliable models of degradation and failure. For example, currently employed chip-scale EM assessment consists of the estimation of a temperature and a current density for each interconnect segment and calculation of MTTF on the basis of the Black equation [5].

$$MTF = Aj^{-n}\exp\{E_a/kT\} \tag{1}$$

Here, j is the current density and E_a is the EM activation energy. Black has determined the value of "n" as equals to 2. However, it is a today's common understanding that "n" depends on residual stress and current density [6], and its value is highly controversial. In the most advanced cases, before the MTTF is calculated, filtration of the immortal segments based on the Blech's "critical product" is performed. This filtration is based on the calculating the product of electrical current density j and the metal line length l for each segment, and the following comparison of the calculated products with the so-called critical product:

$$(j \times l)_{crit} = \frac{\Omega\sigma_{crit}}{eZ\rho}, \tag{2}$$

derived from the well-known "Blech limit" [4], which states that any segment characterized by a smaller current/length product

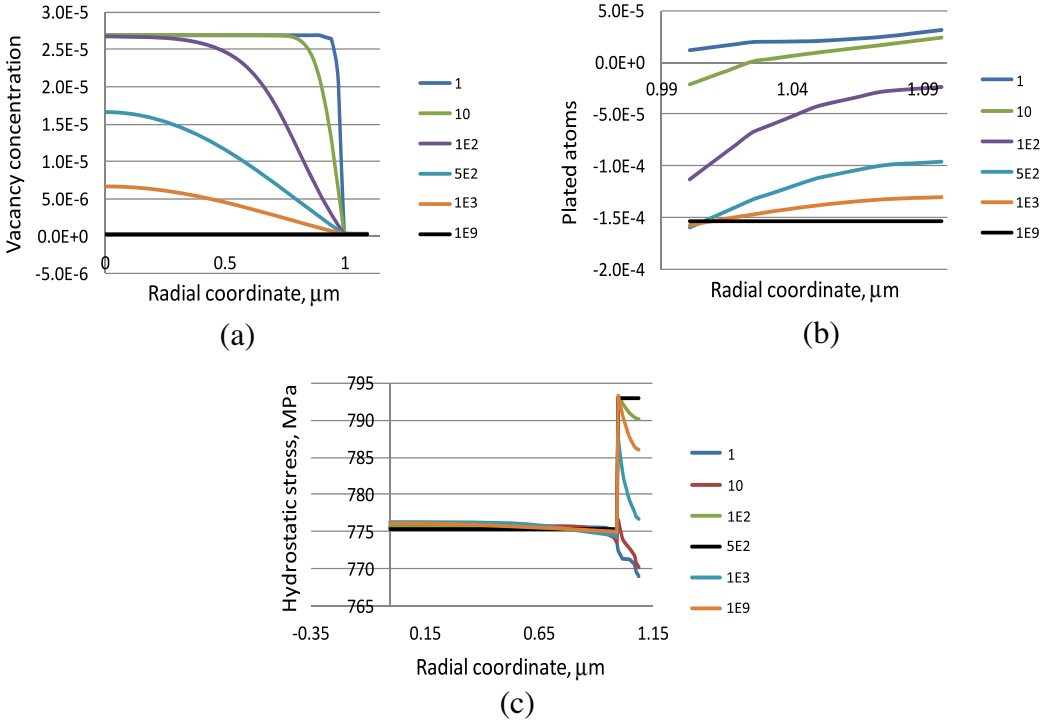


Fig. 4. Kinetics of the radial distribution of the vacancy concentration (a), plated atom concentration (b), and the hydrostatic stress (c). Time is in seconds.

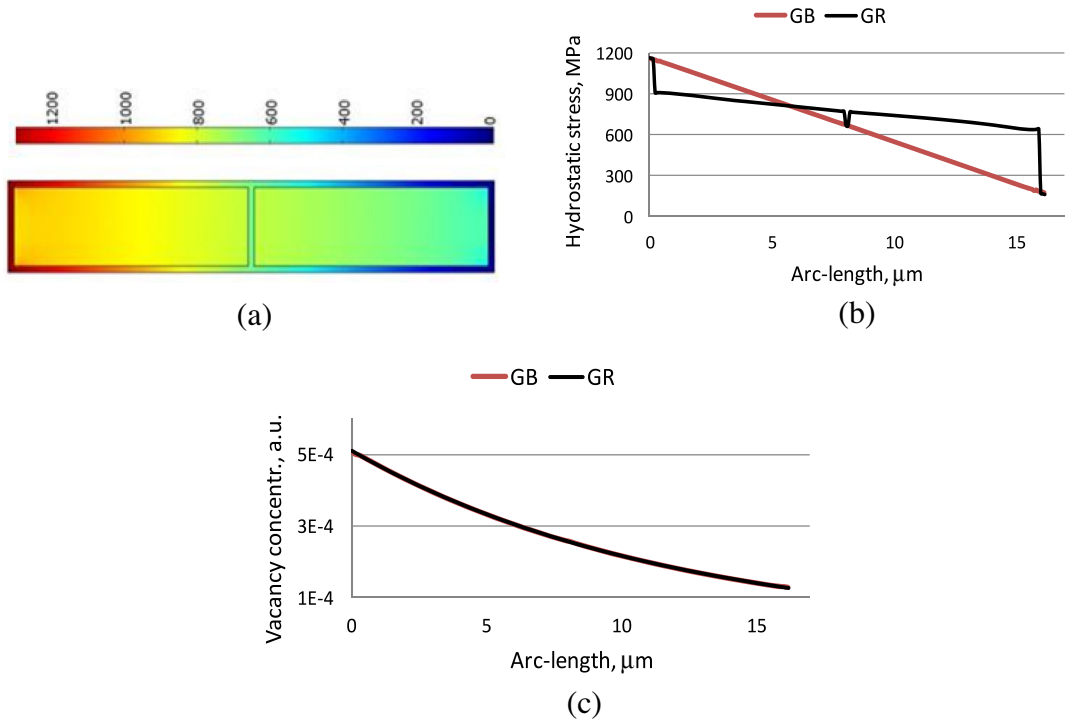


Fig. 5. Hydrostatic stress distribution in two-grain model (a); steady-state distributions of the hydrostatic stress (b), and vacancy concentration (c) in the GB and grain interior.

than a critical one will be immune to the EM failure. The reason is a balance between the EM induced atom flow and the counter flow caused by the stress gradient, which is build-up by the atom/vacancy redistribution. Here, in (2), Ω is the atomic volume; e is the electron charge, eZ is the effective charge of the migrating atoms, ρ is the wire electrical resistivity, k is the Boltzmann's constant; T is the absolute temperature; σ_{crit} is the critical stress needed for the failure nucleation (void/hillock). After sorting-out all immortal segments, the MTTF is calculated for the remaining segments and the minimal MTTF represents the MTTF for the whole die (weakest-link approach). In reality, as it was mentioned above, the situation is not so simple. Since the observed dependency of n and E_a on j and T, the Black equation, that was calibrated at the stressed conditions (elevated j and T), cannot be used for the accurate estimation of MTTF at the use conditions. Strictly speaking, these inter-dependences of n and E_a on j and T undermine the validity of the Black equation by itself [7]. Segment-to-segment variation of the initial (residual) stress makes the "critical product" filtration not as straightforward as well. Now, we need to compare the $(j \times l)$ product for the each interconnect segment with the variable "critical product"

$$(j \times l)^i_{crit} = \frac{\Omega \Delta \sigma^i}{eZ\rho};\ \sigma^i_{init} + \Delta\sigma^i = \sigma_{crit}, \tag{3}$$

where σ^l_{init} is the residual stress existing inside considered interconnect segment before the EM stressing was applied. Hence, the "critical product" is not a constant anymore but a variable, which depends on segment location relatively all possible stress sources. Thus, in the case of the chip-scale EM assessment, in addition to today required assessments of temperature and current density the proper assessment of residual stresses is required. Hence, new, physics-based MTTF compact model, which is free of all discussed flaws related to the Black-Blech formulation, should be developed.

It can be done based on a combination of the FEA-based global/die-scale model and a compact model-based estimation of the post-relaxation stress inside the interconnect segments. The stress relaxation is caused by thermodynamics: the metal lattice vacancies tend to be in the equilibrium with temperature and stress [16]. Change in the state of stress or/and temperature activates a generation/annihilation of vacancies at the grain boundary and a subsequent diffusion of vacancies into the grain bulk. The easiest way to estimate a scale of this relaxation is to consider a single copper grain embedded in a rigid confinement. This grain should be a subject of a change in the interior stress caused, for instance, by a thermal impact. Follow Herring [17], we model a grain as a spherical object of a radius R with the subsurface region of the thickness δ representing a grain boundary, where the generation/annihilation of the vacancy-plated atom pairs occurs. Thus, at the zero stress condition the vacancies are distributed uniformly across the grain with a thermodynamically equilibrium concentration N_0^{ZS}. Both factors, transition to the lower temperature T and generated thermal stress σ_T disrupt the equilibrium in the vacancy concentration. It activates the vacancy migration and the generation/annihilation of the vacancy-plated atom pairs. Plated atoms, which, since their low diffusivity in comparison with the vacancy, are assumed to be immobile, generate a compression while the tensile stress is generated by vacancies [11]. Solution of the standard force balance equation, with the additional terms related to the dilatational strains, caused by vacancies and plated atoms, provides estimates of the new steady state concentrations of vacancies and plated atoms, and the hydrostatic stress values at the grain boundary (GB) and inside the grain interior (GR):

$$\sigma_{Hyd} = \begin{cases} \sigma^{GR}_{Hyd} \approx \sigma_T + \frac{2}{3}\frac{E}{(1-\upsilon)}(N_0^{ZS} - N); & 0 \leqslant r \leqslant R \\ \sigma^{GB}_{Hyd} \approx \sigma_T + \frac{2}{9}\frac{E}{(1-\upsilon)}\left(\frac{R}{\delta}\right)(N_0^{ZS} - N); & R \leqslant r \leqslant R+\delta \end{cases} \tag{4}$$

$$N_{GR} = N_{GB} \approx N_0 e^{\frac{f\Omega\sigma_T}{kT}}\left(1 + \frac{2}{9}\frac{\Omega}{kT}\frac{E}{1-\upsilon}\left(\frac{R}{\delta}\right)\left(N_0^{ZS} - N_0 e^{\frac{f\Omega\sigma_T}{kT}}\right)\right)$$
$$\approx N_0 e^{\frac{f\Omega\sigma_T}{kT}}$$

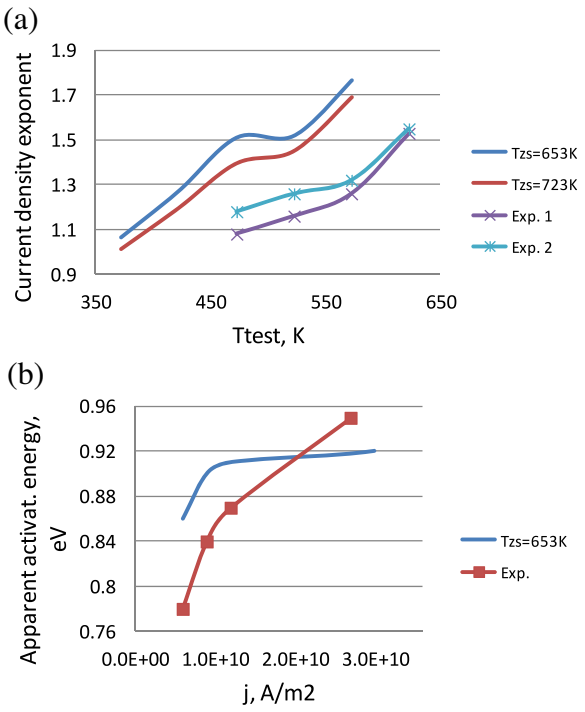


Fig. 6. A comparison of the T_{test} – dependency of n, calculated for two cases: T_{ZS} = 653 K and 723 K, σ_{crit} = 600 MPs, with experimentally determined one: Exp. 1 data from [18] and Exp. 2 from [19], (a); and E_a on j, calculated for T_{ZS} = 653 K and σ_{crit} = 600 MPs against experimental data from [19], (b).

$$M \approx -\frac{1}{3}\left(\frac{R}{\delta}\right)\left(N_0^{ZS} - N_0 e^{\frac{f\Omega\sigma_T}{kT}}\right)$$

Here, $N_0 = \exp\{-E_A/kT\}$ is the equilibrium vacancy concentration at temperature T, N_{GB} and M are the concentrations of vacancies and plated atoms at GB, while N_{GR} is the concentration of vacancies in the grain interior; σ_{Hyd} is the hydrostatic stress; $f = \Omega_v/\Omega$ is the ratio of the volume occupied by a relaxed vacancy to the atomic volume; E and υ are the copper Young's modulus and Poisson factor. Obtained results indicate that the hydrostatic stress is more tensile or less compressive, depending on the nature of σ_T at the GB in comparison with the grain interior. Estimations of the scale of stress relaxation caused by lattice defects equilibration with the temperature and stress provides for GB $\Delta\sigma_{Hyd}^{GB} \approx 20 \div 200\,MPa$ and for grain interior of the order of 100 kPa, which are valid for the ratio (R/δ) of $10^3 \div 10^4$. Excessive tensile stress in the GBs is caused by a larger number of vacancies annihilated with plated atoms due to difference in their equilibrium concentrations at T_{ZS} and at test temperature T in comparison with the number of generated vacancies due to a tension developed by cooling down from T_{ZS} to T. Validation of the derived analytical expressions describing the steady state hydrostatic stress and concentrations of vacancies and plated atoms in the considered grain/grain model was done by comparison with the results of the FEA simulation. Multi-physics SM simulation setup [11,12] was used. Simulation results, shown in Fig. 4, demonstrate evolution of the radial distributions of vacancies, plated atoms and stress. Steady state uniform distributions are achieved at times of order of 10^5 s. Kinetics of this evolution is controlled by diffusion of vacancies from the grain interior to GB where its annihilation with the plated atoms takes place. This process reduces the high initial concentration of vacancies to smaller one corresponding to the test temperature and increases the stress in the GB.

Unidirectional electrical field applied to the interconnect line causes further changes in the hydrostatic stress and concentrations of vacancies and plated atoms. Additional driving force, originated by a momentum exchange between current carriers and lattice atoms, generates an atomic flux flowing toward anode. Results of the FEA simulation of the evolution of hydrostatic stress and concentrations of vacancies and plated atoms demonstrate that the long-term kinetics of EM induced stress relaxation, similarly to considered case of stress-induced relaxation, depends on the vacancy exchange between grain interior and GB. These results were obtained from the simulation of the stress/concentrations evolution kinetics caused by DC stressing in the model system represented by two rectangular grains of length l and diameter $2R$, surrounded by thin GB layer of the thickness δ, and embedded into the rigid confinement (Fig. 5).

Analysis of the GB hydrostatic stress evolution in the vicinity of equilibrium can provide an analytical estimation of the void nucleation time. We assume that a void is nucleated when the hydrostatic stress is reached the critical value of σ_{CR}. It is easy to demonstrate that a void nucleation time is described as:

$$t_{nuc} \approx \tau \frac{kT_{test}}{\Omega B}\frac{1}{\hat{N}_C}\ln\left\{\frac{B\cdot\left(M_T^R - \hat{M}_C\right)}{\sigma_T^R - B\cdot\hat{M}_C - \sigma_{CR}}\right\} \tag{5}$$

Here, $\hat{\sigma}, \hat{N}, \hat{M}$ are the equilibrium values of the hydrostatic stress, vacancy, and plated atom concentrations in GB near the line cathode end, B is the effective bulk elasticity modulus. Parameters σ_T^R, N_T^R, M_T^R represent σ_{Hyd}^{GB} and N_{GB}, M concentrations before an electrical current was applied, described in (4). All considered characteristics are taken for the GB/interface regions since the performed analysis predicts the biggest hydrostatic stress to be developed at the metal line interfaces near the cathode/anode ends. $\hat{N}_C$, $\hat{M}_C$ are the steady state concentrations at the cathode edge: $x = -l/2$. Using Eq. (4) for parameters and the standard formulations for the EM-induced steady state σ_{Hyd}^{GB} and N, M: $\hat{\sigma}(x) = \sigma_T^R - \frac{eZ\rho jx}{2\Omega}$; $\hat{N}(x) = N_T^R e^{-\frac{eZ\rho jx}{2kT}}$; $\hat{M}(x) = M_T^R + \frac{eZ\rho jx}{2\Omega B}$, gives

$$t_{nuc} \approx \tau e^{\frac{E_V}{kT}}\frac{kT}{\Omega B}exp\left\{-\frac{f\Omega\sigma_T}{kT} - \frac{eZ\rho l}{4kT}j\right\}ln\left\{\frac{\frac{eZ\rho l}{4\Omega}j}{\sigma_T + \frac{B}{3}\left(\frac{R}{\delta}\right)e^{\frac{E_V}{kT_{ZS}}} + \frac{eZ\rho l}{4\Omega}j - \sigma_{CR}}\right\};\ \tau \approx \frac{R^2}{D_0}e^{\frac{E_D}{kT}} \tag{6}$$

Effective bulk modulus B, has replaced the multiplier $2E/9(1-\nu)$ from (4), $D = D_0\exp(-E_D/kT)$ – vacancy diffusivity. Thus, the void nucleation time is determined by three major stresses: σ_T, which is layout-dependent combination of the thermal and CPI-induced stresses; residual stress $\sigma_{\mathrm{Res}} = \sigma_T + (B/9)(R/\delta)\exp\{-E_V/kT_{ZS}\}$ existing in the analyzed segment before the electrical stressing is applied; and, finally, the EM-induced stress at the cathode end: $\sigma_{EM} = eZ\rho lj/4\Omega$. Simple analysis of the Eq. (6) reveals that interrelation between three major parameters, namely: j, σ_T and σ_{CR}, governs the void nucleation kinetics. When $\sigma_{CR} > \sigma_R + \sigma_{EM}$, which happens when the condition: $j \times l < 4\Omega(\sigma_{CR} - \sigma_{\mathrm{Res}})/eZ\rho$ is valid, the Eq. (6) yields a divergence of t_{nuc}, which is similar to the Blech condition of immortality [4]. Another situation takes place when $\sigma_{\mathrm{Res}} > \sigma_{CR}$. In this case, the Eq. (6) yields the negative void nucleation time. It is obvious that $\sigma_{\mathrm{Res}} > \sigma_{CR}$ is the condition for stress voiding. Indeed, if the residual stress exceeds the critical stress needed for void nucleation, then void will be nucleated before any electrical stressing is applied. The universal character of the derived formula for the void nucleation time can be further supported by the direct calculation of t_{nuc} for different cur-

rent densities and test temperatures. Representation of the calculated t_{nuc} in the form of the Black equation and extraction of the traditional parameters, which are the current density exponent – n and the apparent activation energy E_a demonstrates that the expected dependencies of n and E_a on T_{test} and j really take place. Comparison of the predicted dependency $n(T_{test})$ with the measurement results [18,19], which is shown in Fig. 6a, demonstrates a reasonable fit. Increase in activation energy E_a calculated under assumption of $E_V \approx 0.7\,\text{eV}$ and $E_D \approx 0.625\,\text{eV}$, caused by increase in the current density (Fig. 6b), corresponds to the experimental observations [19]. This level of fit between the simulated predictions and available experimental data is really encouraging for such idealized model. Model predictability can be improved by introducing a simple calibration step that should tune model parameters until a fit with the measured data is achieved. We hope that further improvement will allow this model to be a fair approximation to the truth.

3. Conclusions

Thus, the scope of the obtained results allows us to conclude that the derived compact model provides a capability for predicting the void nucleation time in interconnect metal segments taking into account the residual stress generated by a combination of thermal treatment and diffusion-induced relaxation. Predictability of the developed model combines together the predictabilities of the Black equation and the Blech limit law, but avoids the errors caused by the across-die variability of the current density exponent n and the apparent activation E_a energy. Combining the accurate estimation of the global stress based on the FEA package/die models with fast estimation of the void nucleation time based on the calibrated version of the developed compact model can create new perspectives for the die-scale SM/EM assessment.

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